



Highway to Renewable Energy Systems

H2RES v1.3 Report (Draft)

Mathematical Formulation and Data Dictionary

Document Type: Technical Document

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1 Scope

This document reconstructs the optimization model implemented in `scripts/Build_model.py`. It also documents the workflow wrapper `scripts/modeling_approaches.py`, the multizone preprocessing logic in `scripts/read_process_data.py`, and the way rolling-horizon execution is embedded procedurally in the wrapper rather than delegated to a separate builder file. It lists all decision variables and all symbols used as parameters/sets/auxiliary terms, then explains the objective and each constraint block.

2 Model Architecture Diagram

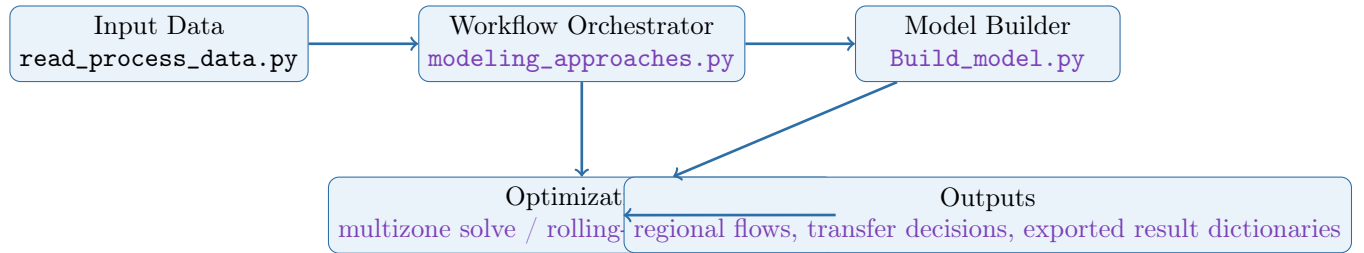


Figure 1: High-level workflow of the H2RES optimization model in the bundled v1.3 multizone structure.

3 General Changes Introduced in v1.1

This summary highlights the main formulation-level differences introduced in the current v1.1 implementation, while the detailed blue additions throughout the rest of the document preserve equation-level traceability.

- The v1.1 model introduces explicit unmet-service variables for electricity, district heating, and general/industrial heat, instead of relying only on the legacy `ceep` slack representation.
- The objective function now includes explicit scarcity and unmet-service penalty terms, so adequacy relaxation is monetized more transparently in the optimization problem.
- Two new scenario controls affect system behavior directly: `ceep_penalty` activates monetary valuation of CEEP, and `fossil_dispatch` can disable dispatch from the coal/gas/diesel subset.
- Thermal and electricity balance closures are extended accordingly, so unmet demand can appear explicitly in the implemented balance equations rather than remaining implicit in a single aggregate slack construct.
- The preprocessing layer now reads the active v1.1 data package from CSV/XLSX sources and builds technology-specific annual CAPEX trajectories for major flexibility technologies, while preserving the same identifier-matching logic across years, periods, units, and fuels.
- The current v1.1 code also uses a reduced time horizon. In the bundled script this appears as `periods = list(range(1,10))`. Treat it as an implementation-specific setting unless it is intentionally part of the study design.

4 General Formulation Changes Introduced in v1.2

This summary isolates the main formulation-level changes introduced in v1.2. Workflow-specific details are retained later in the implementation notes so that the main body remains focused on the optimization structure itself.

- A rolling-horizon counterpart of the core model is now available, so the same system representation can be solved either as a full-horizon problem or through linked subproblems.
- The temporal scope is now parameterized by explicit `periods`, `years`, and `window` controls, so the active optimization horizon is defined transparently rather than being hidden in the builder internals.
- In rolling-horizon mode, investment-carryover equations replace some full-horizon year summations with cumulative investment states plus current-window decisions, so each subproblem depends explicitly on decisions committed in previous windows.
- The v1.2 formulation introduces an auxiliary investment-overflow variable, `extra_cap_inv`, together with an explicit penalty term `extra_investment_cost`, so investments above the nominal annual expansion envelope are allowed but penalized.
- Hydro spillage is now monetized through an active `spill_cost` term in the objective.
- The annual RPS denominator is extended by explicit unmet-electricity service, and the hydrogen-storage lower bound is simplified to a direct non-negativity condition in the implemented v1.2 code.
- Most continuous decision variables are now defined with explicit non-negativity lower bounds, making the domain assumptions of the implemented formulation more transparent.

5 General Formulation Changes Introduced in v1.3

Version 1.3 preserves the sector-coupled logic of earlier releases but restructures the implemented model around an explicitly multizone formulation. The main change is architectural: the same core equation families are now instantiated by region and linked through inter-system flows, national policy layers, and cumulative state updates managed outside the single builder file.

- Most implemented operational and investment variables now carry a leading system index associated with `systems`, so generation, storage, heat, hydrogen, imports, and emissions are all evaluated region by region.
- The preprocessing layer introduces a dedicated system-level workbook, `1_system_data.xlsx`, whose `system_data` and `transfer_limits` sheets define the multizone topology before the technology and demand files are mapped into model sets.
- A new bilateral flow variable, `imports_from`, and an endogenous transfer-capacity investment variable, `transfer_inv`, connect the regional electricity balances through explicit transfer limits and optional expansion decisions.
- Rolling-horizon execution no longer relies on a separate `Build_model_RH.py` builder. Instead, `scripts/modeling_approaches.py` initializes cumulative investment states, solves each window with the single builder, and propagates committed decisions to subsequent windows.

- National policy layers are now explicit through `rps_inv_National`, `carbonLimit_National`, and their annual vectors, so regional and national policy constraints can coexist in the same implementation.

Reading note for the detailed blocks below: to preserve continuity with the v1.0–v1.2 formulation reports, most tables and compact equations still show the base symbol families without the leading system index. In the implemented v1.3 code, the multizone builder prepends that index to most operational, investment, and emissions variables.

6 Data Dictionary Tables (Complete)

6.1 Decision Variables (Complete)

Variable (original model notation)	Type	Unit	Description
$\text{rampup_cost}_{disp_units,t,y}$	Cost auxiliary	EUR	Auxiliary upward ramping cost variable for dispatchable units.
$\text{rampdown_cost}_{disp_units,t,y}$	Cost auxiliary	EUR	Auxiliary downward ramping cost variable for dispatchable units.
$\text{generation}_{u,t,y}$	Operation	MW_e (hourly average)	Hourly electric generation by unit.
$\text{imports}_{t,y}$	Operation	MW_e (hourly average)	Hourly electricity imports.
$\text{imports_from}_{to,from,t,y}$	Operation	MW_e (hourly average)	Bilateral inter-system electricity transfer from origin system from to destination system to .
$\text{transfer_inv}_{to,from,y}$	Investment	MW_e	Endogenous transmission-capacity reinforcement on an existing inter-system link.
$\text{imports_from}_{to,from,t,y}$	Operation	MW_e (hourly average)	Bilateral inter-system electricity transfer from origin system from to destination system to .
$\text{transfer_inv}_{to,from,y}$	Investment	MW_e	Endogenous transmission-capacity reinforcement on an existing inter-system link.
$\text{ceep}_{t,y}$	Operation	MW_e (hourly average)	Curtailement / non-served energy slack variable (ceep).
$\text{noload_elec}_{t,y}$	Operation	MW_e (hourly average)	Explicit unmet-electricity-service variable introduced in the v1.1 implementation.
$\text{cap_inv}_{u,y}$	Investment	MW_e	Annual generation capacity investment.
$\text{extra_cap_inv}_{u,y}$	Investment auxiliary	MW_e	Auxiliary overflow-investment variable used in v1.2 to penalize capacity additions above the nominal annual expansion limit.
CO2_total_y	Emissions	tCO2/year	Annual CO2 emissions from electric generation.
CO2_total_heat_y	Emissions	tCO2/year	Annual CO2 emissions from heat supply.
$\text{CO2_total_industry}_y$	Emissions	tCO2/year	Annual CO2 emissions from industrial fuels.
$\text{cap_inv_FC}_{FC_units,y}$	Investment	MW_e	Annual fuel-cell capacity investment.
$\text{generation_FC}_{FC_units,t,y}$	Operation	MW_e (hourly average)	Hourly electric generation from fuel cells.
$\text{storage_level}_{hydro_storage_units,t,y}$	State variable	MWh_e	Hydro storage level (HDAM/HPHS).
$\text{sto_out_flow}_{hydro_storage_units,t,y}$	Operation	MW_e eq.	Hydro spill / outflow not converted to electricity.
$\text{heat_gen}_{chp_units,t,y}$	Operation	MW_th	Heat output from CHP units.
$\text{noload_heat_dh}_{chp_units,t,y}$	Operation	MW_th	Explicit unmet district-heating service variable used in the v1.1 demand closure.
$\text{noload_heat_ind}_{t,y}$	Operation	MW_th	Explicit unmet general/industrial heat-service variable used in the v1.1 demand closure.
$\text{heat_input}_{chp_units,t,y}$	Operation	MW_th eq.	Electric-to-heat extraction input in CHP coupling.
$\text{heat_sto_level}_{chp_units,t,y}$	Operation	MWh_th	CHP thermal storage state of charge.
$\text{Fuel_ind}_{industry_fuels,t,y}$	Operation	MW_th	Industrial fuel use by fuel type.
$\text{H2tInd}_{t,y}$	Operation	MW_H2	Hydrogen use in industrial high-temperature demand.
$\text{heat_pump_out}_{HeatPump_units,t,y}$	Operation	MW_th	Heat-pump operation or expansion variable by market/segment.
$\text{heat_pump_dh}_{HeatPump_units,chp_units,t,y}$	Operation	MW_th	Heat-pump operation or expansion variable by market/segment.

Variable (original model notation)	Type	Unit	Description
$\text{heat_pump_ind}_{HeatPump_units,t,y}$	Operation	MW_th	Heat-pump operation or expansion variable by market/segment.
$\text{heat_pump_dh_cap}_{HeatPump_units,chp_units,y}$	Investment	MW_th	Heat-pump operation or expansion variable by market/segment.
$\text{heat_pump_ind_cap}_{HeatPump_units,y}$	Investment	MW_th	Heat-pump operation or expansion variable by market/segment.
$\text{cooling_pump_dh}_{HeatPump_units,chp_units,t,y}$	Operation	MW_cool	Cooling output from heat-pump technologies.
$\text{cooling_pump_ind}_{HeatPump_units,t,y}$	Operation	MW_cool	Cooling output from heat-pump technologies.
$\text{boiler_dh_generation}_{thermal_units,chp_units,t,y}$	Operation	MW_th	Thermal boiler heat production variable.
$\text{boiler_ind_generation}_{thermal_units,t,y}$	Operation	MW_th	Thermal boiler heat production variable.
$\text{trad_boiler_ind_cap_inv}_{thermal_units,y}$	Investment	MW_th	Thermal boiler capacity investment variable.
$\text{trad_boiler_dh_cap_inv}_{thermal_units,chp_units,y}$	Investment	MW_th	Thermal boiler capacity investment variable.
$\text{ResToHeat}_{t,y}$	Operation	MW_e (hourly average)	Electricity converted to heat/cooling services (power-to-heat).
$\text{heat_pump_cap}_{HeatPump_units,y}$	Investment	MW_th	Heat-pump operation or expansion variable by market/segment.
$\text{heat_pump_sto_soc}_{HeatPump_units,t,y}$	State variable	MWh_th	Heat-pump operation or expansion variable by market/segment.
$\text{ev_sto_in}_{t,y}$	Operation	MW_e	EV fleet charging/discharging/storage operation variable.
$\text{ev_sto_out}_{t,y}$	Operation	MW_e	EV fleet charging/discharging/storage operation variable.
$\text{ev_sto_soc}_{t,y}$	State variable	MWh_e	Aggregate EV fleet state of charge.
$\text{stat_sto_in}_{Stat_storage_unit,t,y}$	Operation	MW_e	Stationary battery operation/investment variable.
$\text{stat_sto_out}_{Stat_storage_unit,t,y}$	Operation	MW_e	Stationary battery operation/investment variable.
$\text{stat_sto_soc}_{Stat_storage_unit,t,y}$	State variable	MWh_e	Stationary battery state of charge.
$\text{stat_sto_cap_inv}_{Stat_storage_unit,y}$	Investment	MWh_e	Stationary battery operation/investment variable.
$\text{power_to_h2}_{t,y}$	Operation	MW_e	Total electric load used for hydrogen production.
$\text{h2_sto_soc}_{H2_storage_unit,t,y}$	State variable	MWh_H2	Hydrogen storage state of charge.
$\text{h2_sto_out}_{H2_storage_unit,t,y}$	Operation	MW_H2	Hydrogen storage operation/investment variable.
$\text{h2_sto_max_cap}_{H2_storage_unit,y}$	Investment	MWh_H2	Hydrogen storage operation/investment variable.
$\text{h2_electrolysis_max_cap}_{Elec_h2_units,y}$	Investment	MW_H2	Decision variable used in the optimization model.
$\text{generation_H2}_{Elec_h2_units,t,y}$	Operation	MW_H2	Hydrogen production by electrolyzers.
$\text{tot_electric_load}_{t,y}$	Auxiliary	MW_e	Decision variable used in the optimization model.
$\text{max_cap_installed}_y$	Auxiliary	MW_e	Decision variable used in the optimization model.

6.2 Parameters, Sets, and Auxiliary Terms (Complete)

Parameter / set (original model notation)	Category	Unit	Description
$\text{Boiler_decom_rate}_{thermal_units}$	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
$\text{Boiler_decom_start_new}_{thermal_units}$	Technical	year	Thermal boiler module parameter.
$\text{CHPMaxHeat}_{chp_units}$	Technical	MW	CHP coupling / thermal storage parameter.
$\text{CHPPowerLossFactor}_{chp_units}$	Technical	p.u.	CHP coupling / thermal storage parameter.
$\text{CHPPowerToHeat}_{chp_units}$	Technical	p.u.	CHP coupling / thermal storage parameter.
CO2_cost_industry	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.

Parameter / set (original model notation)	Category	Unit	Description
CO2_emissions_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
ceep_cost	Objective auxiliary	EUR	Conditional scarcity-cost term activated when <code>ceep_penalty=True</code> .
no_load_cost	Objective auxiliary	EUR	Penalty term for the explicit electricity and general-heat unmet-service variables in v1.1.
no_load_cost_dh	Objective auxiliary	EUR	Penalty term for explicit unmet district-heating service in v1.1.
spill_cost	Objective auxiliary	EUR	Active hydro-spillage penalty term in the v1.2 objective implementation.
extra_investment_cost	Objective auxiliary	EUR	Penalty term applied to <code>extra_cap_inv</code> in the v1.2 implementation.
CO2_factor _u	Technical	tCO2/MWh	CO2 intensity factor.
CO2_limit _y	Policy	tCO2/year	Policy parameter by model year.
CO2_price _y	Policy	EUR/tCO2	Policy parameter by model year.
Elec_h2_units	Set	-	Index set/subset used in constraints and objective terms.
FC_eff _{FC_units}	Technical	p.u.	Hydrogen and fuel-cell module parameter.
FC_init_cap _{FC_units}	Technical	MW	Hydrogen and fuel-cell module parameter.
FC_max_cap _{FC_units}	Technical	MW	Hydrogen and fuel-cell module parameter.
FC_units	Set	-	Index set/subset used in constraints and objective terms.
FC_var_cost _{FC_units}	Technical	EUR/MWh	Hydrogen and fuel-cell module parameter.
Fossil_hydro	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
Fuel_CO2 _{industry_fuels}	Technical	tCO2/MWh	CO2 intensity factor.
H2_demand _{demand_sectors,t,y}	Demand	MW_H2	Exogenous demand trajectory by market/sector and time.
H2_eff _{Elec_h2_units}	Technical	p.u.	Hydrogen and fuel-cell module parameter.
H2_init_cap _{Elec_h2_units}	Technical	MW	Hydrogen and fuel-cell module parameter.
H2_max_cap _{Elec_h2_units}	Technical	MW	Hydrogen and fuel-cell module parameter.
H2_sto_init _{H2_storage_unit}	Technical	MWh	Hydrogen and fuel-cell module parameter.
H2_sto_max_cap _{H2_storage_unit}	Technical	MWh	Hydrogen and fuel-cell module parameter.
H2_storage_unit	Technical	model units	Hydrogen and fuel-cell module parameter.
H2_var_cost _{Elec_h2_units}	Technical	EUR/MWh	Hydrogen and fuel-cell module parameter.
HP_Boiler_dh_init_cap _{HeatPump_units,chp_units}	Technical	MW	Heat pump / heat system parameter.
HP_COP_period _{HeatPump_units,t,y}	Technical	p.u.	Heat pump / heat system parameter.
HP_Decom_ind _{HeatPump_units,y}	Technical	p.u.	Heat pump / heat system parameter.
HP_decom_rate _{HeatPump_units}	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
HP_decom_start_new _{HeatPump_units}	Technical	year	Heat pump / heat system parameter.
HP_ind_init_cap _{HeatPump_units}	Technical	MW	Heat pump / heat system parameter.
HP_max_cap _{HeatPump_units}	Technical	MW	Heat pump / heat system parameter.
HP_sto_init _{HeatPump_units}	Technical	MWh	Heat pump / heat system parameter.
HP_var_cost _{HeatPump_units}	Technical	model units	Heat pump / heat system parameter.
HeatPump_units	Set	-	Index set/subset used in constraints and objective terms.
Imp_price_inc _y	Economic	% per year	Exogenous economic parameter.
Import_Price _{t,y}	Economic	EUR/MWh_e	Exogenous economic parameter.

Parameter / set (original model notation)	Category	Unit	Description
systems	Set	-	Top-level set of modeled regions or zones in the multizone v1.3 implementation.
transfer_limits _{to,from}	Technical	MW_e	Existing bilateral transfer-capacity limit read from the transfer_limits sheet in 1_system_data.xlsx .
Transmission_Inv	Scenario flag	bool	Boolean switch that activates or blocks endogenous transmission-capacity expansion.
rps_inv_National	Scenario flag	bool	Boolean switch that activates the national renewable-share policy layer.
carbonLimit_National	Scenario flag	bool	Boolean switch that activates the national CO2-cap policy layer.
rps_National _y	Policy	p.u.	Annual national renewable-share target applied across the sum of all systems.
CO2_Limit_National _y	Policy	tCO2/year	Annual national CO2 cap applied to the aggregate of all regional emissions.
systems	Set	-	Top-level set of modeled regions or zones in the multizone v1.3 implementation.
transfer_limits _{to,from}	Technical	MW_e	Existing bilateral transfer-capacity limit read from the transfer_limits sheet in 1_system_data.xlsx .
Transmission_Inv	Scenario flag	bool	Boolean switch that activates or blocks endogenous transmission-capacity expansion.
rps_inv_National	Scenario flag	bool	Boolean switch that activates the national renewable-share policy layer.
carbonLimit_National	Scenario flag	bool	Boolean switch that activates the national CO2-cap policy layer.
rps_National _y	Policy	p.u.	Annual national renewable-share target applied across the sum of all systems.
CO2_Limit_National _y	Policy	tCO2/year	Annual national CO2 cap applied to the aggregate of all regional emissions.
Industry_HT_demand _{t,y}	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
NPV	Economic	p.u.	Exogenous economic parameter.
NoResToHeatInv	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
ResToHeat_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
ST0Capacity_heat _{chp_units}	Technical	MWh	CHP coupling / thermal storage parameter.
StaSto-Decom_ind _y	Technical	p.u.	Exogenous parameter or helper alias used in constraints/objective.
Sta_sto_decom_rate _{Stat_storage_unit}	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
Sta_sto_decom_start_new _{Stat_storage_unit}	Technical	year	Stationary storage module parameter.
Sta_sto_eff _{Stat_storage_unit}	Technical	p.u.	Stationary storage module parameter.
Sta_sto_init_cap _{Stat_storage_unit}	Technical	MWh	Stationary storage module parameter.
Sta_sto_max_cap _{Stat_storage_unit}	Technical	MWh	Stationary storage module parameter.
Sta_sto_var_cost _{Stat_storage_unit}	Technical	model units	Stationary storage module parameter.
Stat_storage_unit	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
TechChange	Technical	p.u.	Capacity expansion/decommissioning trajectory parameter.
Thermal_CO2 _{thermal_units}	Technical	tCO2/MWh	CO2 intensity factor.
Thermal_Decom_DH _{thermal_units,chp_units,y}	Technical	p.u.	Thermal boiler module parameter.
Thermal_Decom_ind _{thermal_units,y}	Technical	p.u.	Thermal boiler module parameter.
Thermal_ind_init_cap _{thermal_units}	Technical	MW	Thermal boiler module parameter.
Thermal_max_cap _{thermal_units}	Technical	MW	Thermal boiler module parameter.
V2G_cost _{t,y}	Economic	EUR/MWh_e	Exogenous economic parameter.
aval_factor_plant _{u,t,y}	Technical	p.u.	Hydro inflow/efficiency/availability parameter.
biomass_units	Set	-	Index set/subset used in constraints and objective terms.

Parameter / set (original model notation)	Category	Unit	Description
boiler_gen_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
cap_factor_u	Technical	p.u.	Technical/economic parameter for generation technologies.
cap_inv_cost_{u,y}	Technical	EUR/MW	Technical/economic parameter for generation technologies.
carbonLimit	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
ceep_demand_y	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
ceep_limit	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
ceep_penalty	Scenario flag	bool	Boolean switch that activates/deactivates monetization of CEEP in the objective.
ceep_value	Policy	EUR/MWh_e	Scarcity-value parameter applied to CEEP when the penalty switch is active.
ceep_parameter_y	Policy	p.u.	Upper bound on annual ceep as share of annual demand.
chp_units	Set	-	Index set/subset used in constraints and objective terms.
cooling_demand_{demand_sectors,t,y}	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
decom_rate_u	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
decom_start_new_u	Technical	year	Capacity expansion/decommissioning trajectory parameter.
decom_start_old_u	Technical	year	Capacity expansion/decommissioning trajectory parameter.
demand_{demand_sectors,t,y}	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
demand_sectors	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
disp_units	Set	-	Index set/subset used in constraints and objective terms.
evToPowert_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
ev_Demand_y	Technical	MW-equivalent (hourly)	EV/V2G module parameter.
ev_Grid_P_max_y	Technical	MW	EV/V2G module parameter.
ev_Grid_eff_y	Technical	p.u.	EV/V2G module parameter.
ev_availability_{t,y}	Technical	p.u.	EV/V2G module parameter.
ev_sto_max_y	Technical	MWh	EV/V2G module parameter.
ev_sto_min_y	Technical	p.u.	EV/V2G module parameter.
ev_transp_load_{t,y}	Technical	MW-equivalent (hourly)	EV/V2G module parameter.
exports_{t,y}	Technical	MW_e	Interconnection/import-export parameter by period.
fossil_dispatch	Scenario flag	bool	Boolean switch that can force the coal/gas/diesel subset out of dispatch.
Rolling_Horizon	Scenario flag	bool	Boolean switch that routes the v1.2 workflow to the rolling-horizon solution approach.
window	Technical configuration	years	Number of planning years optimized simultaneously in each rolling-horizon subproblem.
LostLoad_Cost	Economic	EUR/MWh	Penalty coefficient used for explicit unmet-service variables in the v1.1 implementation.
fossil_units	Set	-	Index set/subset used in constraints and objective terms.
coal_gas_diesel_units	Set	-	Subset of fossil units affected by the optional fossil_dispatch shutdown logic in v1.1.
fuel_share_max_{t,y,industry_fuels}	Technical	p.u.	Exogenous parameter or helper alias used in constraints/objective.
h2_CAPEX_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.

Parameter / set (original model notation)	Category	Unit	Description
h2_OPEX_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
h2_storage_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
hdam_efficiency _u	Technical	p.u.	Hydro inflow/efficiency/availability parameter.
hdam_inflow _{u,t,y}	Technical	inflow p.u.	Hydro inflow/efficiency/availability parameter.
hdam_max_storage _u	Technical	MWh	Hydro inflow/efficiency/availability parameter.
hdam_units	Set	-	Index set/subset used in constraints and objective terms.
heat_demand _{chp_units,t,y}	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
heat_pump_cap_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
heat_var_cost _{HeatPump_units}	Technical	EUR/MWh	Heat pump / heat system parameter.
hphs_efficiency _u	Technical	p.u.	Hydro inflow/efficiency/availability parameter.
hphs_inflow _{u,t,y}	Technical	inflow p.u.	Hydro inflow/efficiency/availability parameter.
hphs_max_storage _u	Technical	MWh	Hydro inflow/efficiency/availability parameter.
hphs_units	Set	-	Index set/subset used in constraints and objective terms.
hror_units	Set	-	Index set/subset used in constraints and objective terms.
hydro_storage	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
hydro_storage_units	Set	-	Index set/subset used in constraints and objective terms.
hydro_units	Set	-	Index set/subset used in constraints and objective terms.
import_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
import_ntc _t	Technical	MW_e	Interconnection/import-export parameter by period.
industry_fuel_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
industry_fuels	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
industry_fuels_price _{industry_fuels,t,y}	Technical	EUR/MWh_fuel	Exogenous parameter or helper alias used in constraints/objective.
inv_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
inv_cost_FuelCell	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
inv_cost_boiler	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
inv_cost_fossil	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
max_inv_period _u	Technical	MW	Capacity expansion/decommissioning trajectory parameter.
max_load _u	Technical	MW_e	Technical/economic parameter for generation technologies.
nondisp_units	Set	-	Index set/subset used in constraints and objective terms.
per_mw	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
per_mw_FC	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
periods	Set	-	Index set/subset used in constraints and objective terms.
ramp_cost _u	Technical	EUR/MW	Ramping capability/cost parameter.
ramp_down_cost_var	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.

Parameter / set (original model notation)	Category	Unit	Description
<code>ramp_down_rate_u</code>	Technical	p.u. per hour/year	Ramping capability/cost parameter.
<code>ramp_up_cost_var</code>	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
<code>ramp_up_rate_u</code>	Technical	p.u. per hour/year	Ramping capability/cost parameter.
<code>rps_inv</code>	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
<code>statCapInvCost</code>	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
<code>statToPowert_cost</code>	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
<code>Elec_cap_costs_years_{Elec_h2_units,y}</code>	Technical	EUR/MW	Annual electrolyzer CAPEX trajectory read from the v1.1 technology-cost workbook.
<code>FC_cap_costs_years_{FC_units,y}</code>	Technical	EUR/MW	Annual fuel-cell CAPEX trajectory read from the v1.1 technology-cost workbook.
<code>HP_cap_costs_years_{HeatPump_units,y}</code>	Technical	EUR/MW	Annual heat-pump CAPEX trajectory read from the v1.1 technology-cost workbook.
<code>boiler_cap_costs_years_{thermal_units,y}</code>	Technical	EUR/MW	Annual thermal-boiler CAPEX trajectory read from the v1.1 technology-cost workbook.
<code>H2sto_costs_years_{H2_storage_unit,y}</code>	Technical	EUR/MWh	Annual hydrogen-storage CAPEX trajectory read from the v1.1 technology-cost workbook.
<code>battery_costs_years_{Stat_storage_unit,y}</code>	Technical	EUR/MWh	Annual stationary-battery CAPEX trajectory read from the v1.1 technology-cost workbook.
<code>thermal_units</code>	Set	-	Index set/subset used in constraints and objective terms.
<code>units</code>	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
<code>var_cost_u</code>	Technical	EUR/MWh	Technical/economic parameter for generation technologies.
<code>years</code>	Set	-	Index set/subset used in constraints and objective terms.

Unit note: The model is hourly. Several power and energy quantities are numerically linked within each hour (MW average vs MWh per hour). Units above follow the most interpretable convention for each symbol.

7 Formulation by Equation Blocks and Their Role

Compact notation: $u \in \text{units}$, $t \in \text{periods}$, $y \in \text{years}$, $G_{u,t,y} = \text{generation}_{u,t,y}$, $I_{t,y} = \text{imports}_{t,y}$, $C_{t,y} = \text{ceep}_{t,y}$.

7.1 1) Objective Function

Description: This block aggregates all discounted annual cost components (operation, investment, flexibility, hydrogen, heat, and policy-related terms) into a single scalar objective Z .

$$\begin{aligned} \min Z = & \text{per_mw} + \text{import_cost} + \text{inv_cost} + \text{inv_cost_fossil} \\ & + \text{ramp_up_cost_var} + \text{ramp_down_cost_var} + \text{heat_pump_cap_cost} \\ & + \text{ResToHeat_cost} + \text{boiler_gen_cost} + \text{inv_cost_boiler} + \text{CO2_emissions_cost} \\ & + \text{evToPowert_cost} + \text{statToPowert_cost} + \text{statCapInvCost} \\ & + \text{h2_storage_cost} + \text{h2_CAPEX_cost} + \text{h2_OPEX_cost} \\ & + \text{ceep_cost} + \text{no_load_cost} + \text{no_load_cost_dh} \\ & + \text{inv_cost_FuelCell} + \text{per_mw_FC} + \text{industry_fuel_cost} \end{aligned} \quad (1)$$

Role: minimizes total discounted system cost across electricity, heat/cooling, hydrogen, transport flexibility, and investment decisions. In the current v1.1 code, this role is extended by explicit scarcity penalties for CEEP and for unmet electricity/heat service. In v1.2, the objective also penalizes hydro spillage and investment above the nominal annual expansion envelope.

7.2 2) Policy and Scenario Activation Blocks

Description: These logical switches activate or deactivate optional model modules through equivalent algebraic constraints, enabling scenario analysis without changing the model core.

$$\text{NoResToHeatInv=True} \Rightarrow \text{heat_pump_cap} = 0 \quad (2)$$

$$\text{NoResToHeatInv=True} \Rightarrow \text{ResToHeat} = 0 \quad (3)$$

$$\text{NoResToHeatInv=True} \Rightarrow \text{heat_pump_out} = 0 \quad (4)$$

$$\text{rps_inv=True} \Rightarrow \text{RPS constraint is active} \quad (5)$$

$$\text{carbonLimit=True} \Rightarrow \text{annual CO2 cap is active} \quad (6)$$

$$\text{PrimaryReserve/SecondaryReserve=True} \Rightarrow \text{reserve modules are active} \quad (7)$$

$$\text{ceep_penalty=True} \Rightarrow \text{CEEP is monetized in the objective} \quad (8)$$

$$\text{fossil_dispatch=False} \Rightarrow \text{generation}_{u,t,y} = 0, u \in \text{coal_gas_diesel_units} \quad (9)$$

Role: allows policy/scenario switching without changing the core model structure.

7.3 3) Primary Reserve (if active)

Description: The primary reserve module enforces hourly up/down reserve requirements and limits reserve provision using unit headroom/footroom and storage/flexibility feasibility conditions. **Role:** enforces short-term balancing capability.

7.4 4) Secondary Reserve (if active)

Description: This block mirrors the primary reserve logic with dedicated variables and constraints for secondary reserve products. **Role:** enforces additional balancing margin.

7.5 5) Industrial Heat Demand and Fuel Switching

Description: The equations below set a fuel-share upper bound per fuel and period, then close the industrial heat balance with endogenous hydrogen substitution.

$$\text{Fuel_ind}_{f,t,y} \leq \text{fuel_share_max}_{t,y,f} \cdot (\text{Industry_HT_demand}_{t,y} - \text{H2tInd}_{t,y}) \quad (10)$$

$$\sum_f \text{Fuel_ind}_{f,t,y} + \text{H2tInd}_{t,y} = \text{Industry_HT_demand}_{t,y} \quad (11)$$

Role: satisfies industrial thermal demand while allowing endogenous fuel mix with hydrogen substitution.

7.6 6) Hydrogen Production, Storage, and Demand Balance

Description: This block links electricity use to hydrogen production via efficiency, limits hydrogen conversion capacity, updates hydrogen storage SOC, and balances total hydrogen withdrawals with final uses.

$$\sum_{e \in \text{Elec_h2_units}} \frac{\text{generation_H2}_{e,t,y}}{\text{H2_eff}_e} = \text{power_to_h2}_{t,y} \quad (12)$$

$$\text{generation_H2}_{e,t,y} \leq \text{H2_init_cap}_e + \sum_{\tau \leq y} \text{h2_electrolysis_max_cap}_{e,\tau} \quad (13)$$

$$\text{h2_sto_soc}_{s,t,y} = \text{h2_sto_soc}_{s,t-1,y} - \text{h2_sto_out}_{s,t,y} + \sum_e \text{generation_H2}_{e,t,y} \quad (14)$$

with SOC bounds and total demand balance:

$$\sum_s \text{h2_sto_out}_{s,t,y} = \sum_q \text{H2_demand}_{q,t,y} + \sum_c \frac{\text{generation_FC}_{c,t,y}}{\text{FC_eff}_c} + \text{H2tInd}_{t,y} \quad (15)$$

Role: ensures physical consistency of the hydrogen subsystem.

7.7 7) EV and Stationary Storage Dynamics

Description: These equations track EV and stationary storage SOC trajectories over time with efficiency-adjusted charge/discharge flows, while enforcing SOC bounds, power limits, investment-based capacity limits, and end-of-horizon conditions.

7.7.1 7.1 EV Fleet Storage

Compact notation:

$$\begin{aligned}
 E_{t,y}^{ev} &\equiv \text{ev_sto_soc}_{t,y}, & P_{t,y}^{ev,in} &\equiv \text{ev_sto_in}_{t,y}, & P_{t,y}^{ev,out} &\equiv \text{ev_sto_out}_{t,y}, \\
 \eta^{ev} &\equiv \text{ev_Grid_eff}, & \bar{E}_y^{ev} &\equiv \text{ev_sto_max}_y, & \alpha^{ev} &\equiv \text{ev_sto_min}, \\
 \bar{P}_y^{ev} &\equiv \text{ev_Grid_P_max}_y, & a_{t,y}^{ev} &\equiv \text{ev_availability}_{t,y}, & L_{t,y}^{ev} \cdot D_y^{ev} &\equiv \text{ev_transp_load}_{t,y} \cdot \text{ev_Demand}_y.
 \end{aligned}$$

$$E_{1,y}^{ev} = -L_{1,y}^{ev} \cdot D_y^{ev} - P_{1,y}^{ev,out} + \eta^{ev} \cdot P_{1,y}^{ev,in} \quad (16)$$

$$E_{t,y}^{ev} = E_{t-1,y}^{ev} - L_{t,y}^{ev} \cdot D_y^{ev} - P_{t,y}^{ev,out} + \eta^{ev} \cdot P_{t,y}^{ev,in} \quad (17)$$

$$\alpha^{ev} \cdot \bar{E}_y^{ev} \leq E_{t,y}^{ev} \leq \bar{E}_y^{ev} \quad (18)$$

$$P_{t,y}^{ev,out} \leq \bar{P}_y^{ev} \cdot a_{t,y}^{ev} \quad (19)$$

$$P_{t,y}^{ev,in} \leq \bar{P}_y^{ev} \cdot a_{t,y}^{ev} \quad (20)$$

7.7.2 7.2 Stationary Battery Storage

Compact notation:

$$\begin{aligned}
 E_{b,t,y}^{st} &\equiv \text{stat_sto_soc}_{b,t,y}, & P_{b,t,y}^{st,in} &\equiv \text{stat_sto_in}_{b,t,y}, \\
 P_{b,t,y}^{st,out} &\equiv \text{stat_sto_out}_{b,t,y}, & \eta_b^{st} &\equiv \text{Sta_sto_eff}_b, \\
 \bar{E}_{b,y}^{st} &\equiv \text{Sta_sto_init_cap}_b \cdot \text{StaSto_Decom_ind}_y + \sum_{\tau \leq y} \text{stat_sto_cap_inv}_{b,\tau} \cdot \text{StaStoDecom}_{b,\tau,y}.
 \end{aligned}$$

$$E_{b,1,y}^{st} = -P_{b,1,y}^{st,out} + \eta_b^{st} \cdot P_{b,1,y}^{st,in} \quad (21)$$

$$E_{b,t,y}^{st} = E_{b,t-1,y}^{st} - P_{b,t,y}^{st,out} + \eta_b^{st} \cdot P_{b,t,y}^{st,in} \quad (22)$$

$$E_{b,t,y}^{st} \leq \bar{E}_{b,y}^{st} \quad (23)$$

$$\sum_y \text{stat_sto_cap_inv}_{b,y} \leq \text{Sta_sto_max_cap}_b \quad (24)$$

$$E_{b,T,y}^{st} \leq 0.2 \text{Sta_sto_max_cap}_b \quad (25)$$

Role: shifts load and supports balancing through electrochemical storage.

7.8 8) Generation Limits by Technology Type

Description: The block first defines effective available capacity from surviving existing assets plus surviving new investments, then applies technology-specific dispatch rules and annual investment ceilings.

$$\text{CapEff}_{u,y} = \text{max_load}_u \cdot \text{DecomOld}_{u,y} + \sum_{\tau \leq y} \text{cap_inv}_{u,\tau} \cdot \text{DecomNew}_{u,\tau,y} \quad (26)$$

$$\text{generation}_{u,t,y} \leq \text{cap_factor}_u \cdot \text{CapEff}_{u,y}, \quad u \in \text{fossil_units} \cup \text{hphs_units} \cup \text{hdam_units} \quad (27)$$

$$\text{generation}_{u,t,y} = \text{aval_factor_plant}_{u,t,y} \cdot \text{CapEff}_{u,y}, \quad u \in \text{nondisp_units} \cup \text{hror_units} \quad (28)$$

$$\text{cap_inv}_{u,y} \leq \text{max_inv_period}_u \quad (29)$$

$$\text{extra_cap_inv}_{u,y} \geq \text{cap_inv}_{u,y} - 0.8 \cdot \text{max_inv_period}_u \quad (30)$$

Role: enforces physical and planning feasibility of generation. In the current v1.2 implementation, capacity additions can exceed the nominal annual envelope only through the auxiliary overflow variable `extra_cap_inv`, which is then penalized in the objective. This mechanism is especially relevant in the rolling-horizon setting because it discourages front-loading of capacity expansion in early windows while still preserving feasibility.

7.9 9) Electricity Balance and CEPP Bound

Description: This block enforces hourly electricity adequacy by balancing injections (generation, imports, and storage discharge) with withdrawals (demand, charging, and sector-coupling electric loads). It also bounds annual `ceep` as a fraction of yearly electric-energy requirements. *Compact notation:*

$$\begin{aligned} \mathcal{U}^{el} &\equiv \text{fossil_units} \cup \text{hphs_units} \cup \text{hdam_units} \cup \text{hror_units} \cup \text{nondisp_units}, \\ G_{t,y}^{tot} &\equiv \text{GenTot}_{t,y}, \quad G_{u,t,y} \equiv \text{generation}_{u,t,y}, \quad G_{c,t,y}^{FC} \equiv \text{generation_FC}_{c,t,y}, \\ I_{t,y} &\equiv \text{imports}_{t,y}, \quad D_{s,t,y} \equiv \text{demand}_{s,t,y}, \quad C_{t,y} \equiv \text{ceep}_{t,y}, \\ P_{t,y}^{ev,in} &\equiv \text{ev_sto_in}_{t,y}, \quad P_{t,y}^{ev,out} \equiv \text{ev_sto_out}_{t,y}, \quad \eta^{ev} \equiv \text{ev_Grid_eff}, \\ P_{b,t,y}^{st,in} &\equiv \text{stat_sto_in}_{b,t,y}, \quad P_{b,t,y}^{st,out} \equiv \text{stat_sto_out}_{b,t,y}, \quad \eta_b^{st} \equiv \text{Sta_sto_eff}_b, \\ H_{t,y} &\equiv \text{power_to_h2}_{t,y}, \quad R_{t,y} \equiv \text{ResToHeat}_{t,y}, \quad \bar{\alpha}^{ceep} \equiv \text{ceep_parameter}, \\ N_{t,y}^{el} &\equiv \text{noload_elect}_{t,y}. \end{aligned}$$

$$G_{t,y}^{tot} = \sum_{u \in \mathcal{U}^{el}} G_{u,t,y} + \sum_{c \in \text{FC_units}} G_{c,t,y}^{FC} \quad (31)$$

$$G_{t,y}^{tot} + I_{t,y} + \eta^{ev} \cdot P_{t,y}^{ev,out} + \sum_b \eta_b^{st} \cdot P_{b,t,y}^{st,out} + N_{t,y}^{el} = \sum_s D_{s,t,y} + \sum_b P_{b,t,y}^{st,in} + P_{t,y}^{ev,in} + H_{t,y} + R_{t,y} + C_{t,y} \quad (32)$$

$$\sum_t C_{t,y} \leq \bar{\alpha}^{ceep} \cdot \left(\sum_{s,t} D_{s,t,y} + \sum_{b,t} P_{b,t,y}^{st,in} + \sum_t R_{t,y} + \sum_t H_{t,y} + \sum_t P_{t,y}^{ev,in} \right) \quad (33)$$

Role: central electricity adequacy condition of the integrated system, plus an explicit annual quality/reliability bound through limited `ceep`. In the implemented v1.1 code, adequacy can also be relaxed through the explicit variable `no_load_elec`, and the annual CEEP denominator is evaluated against net storage charging rather than gross charging alone.

7.10 10) Heat/Cooling Block (HP, Boilers, CHP)

Description: This block maps electricity-to-heat conversion via COP relations, enforces heat-pump and boiler operating limits, captures CHP heat-power coupling, updates thermal storage states, and closes hourly heat/cooling demand balances. *Compact notation:*

$$\begin{aligned} Q_{h,m,t,y}^{dh} &\equiv \text{heat_pump_dh}_{h,m,t,y}, & Q_{h,m,t,y}^{cdh} &\equiv \text{cooling_pump_dh}_{h,m,t,y}, \\ Q_{h,t,y}^{ind} &\equiv \text{heat_pump_ind}_{h,t,y}, & Q_{h,t,y}^{cind} &\equiv \text{cooling_pump_ind}_{h,t,y}, \\ Q_{h,t,y}^{out} &\equiv \text{heat_pump_out}_{h,t,y}, & E_{h,t,y}^{hp} &\equiv \text{heat_pump_sto_soc}_{h,t,y}, \\ COP_{h,t,y} &\equiv \text{HP_COP_period}_{h,t,y}, & R_{t,y} &\equiv \text{ResToHeat}_{t,y}, \\ \bar{Q}_{h,m,y}^{dh} &\equiv \text{HPDHCapEff}_{h,m,y}, & \bar{Q}_{h,y}^{ind} &\equiv \text{HPINDCapEff}_{h,y}, \\ B_{b,m,t,y}^{dh} &\equiv \text{boiler_dh_generation}_{b,m,t,y}, & B_{b,t,y}^{ind} &\equiv \text{boiler_ind_generation}_{b,t,y}, \\ \bar{B}_{b,m,y}^{dh} &\equiv \text{BoilerDHCapEff}_{b,m,y}, & \bar{B}_{b,y}^{ind} &\equiv \text{BoilerIndCapEff}_{b,y}, \\ H_{m,t,y}^{in} &\equiv \text{heat_input}_{m,t,y}, & G_{m,t,y} &\equiv \text{generation}_{m,t,y}, \\ \kappa_m^{ht} &\equiv \text{CHPPowerToHeat}_m, & \kappa_m^{loss} &\equiv \text{CHPPowerLossFactor}_m, \\ \bar{G}_m &\equiv \text{max_load}_m \cdot \text{cap_factor}_m, & S_{m,t,y} &\equiv \text{heat_sto_level}_{m,t,y}, \\ Q_{m,t,y}^{gen} &\equiv \text{heat_gen}_{m,t,y}, & D_{m,t,y}^{heat} &\equiv \text{heat_demand}_{m,t,y}, \\ D_{t,y}^{heat,g} &\equiv \text{heat_demand}_{g^{heat},t,y}, & D_{t,y}^{cool,g} &\equiv \text{cooling_demand}_{g^{cool},t,y}, \\ N_{m,t,y}^{dh} &\equiv \text{no_load_heat_dh}_{m,t,y}, & N_{t,y}^{ind} &\equiv \text{no_load_heat_ind}_{t,y}, \\ g^{heat} &\equiv \text{general_demand}, & g^{cool} &\equiv \text{general_cooling}. \end{aligned}$$

$$\sum_{h,m} \frac{Q_{h,m,t,y}^{dh} + Q_{h,m,t,y}^{cdh}}{COP_{h,t,y}} + \sum_h \frac{Q_{h,t,y}^{ind}}{COP_{h,t,y}} = R_{t,y} \quad (34)$$

$$Q_{h,m,t,y}^{dh} + Q_{h,m,t,y}^{cdh} \leq \bar{Q}_{h,m,y}^{dh} \quad (35)$$

$$Q_{h,t,y}^{ind} \leq \bar{Q}_{h,y}^{ind} \quad (36)$$

$$E_{h,t,y}^{hp} = E_{h,t-1,y}^{hp} + Q_{h,t,y}^{ind} - Q_{h,t,y}^{out} - Q_{h,t,y}^{cind} \quad (37)$$

$$B_{b,m,t,y}^{dh} \leq \bar{B}_{b,m,y}^{dh} \quad (38)$$

$$B_{b,t,y}^{ind} \leq \bar{B}_{b,y}^{ind} \quad (39)$$

$$H_{m,t,y}^{in} \cdot \kappa_m^{ht} \leq G_{m,t,y} \quad (40)$$

$$G_{m,t,y} \leq \bar{G}_m - H_{m,t,y}^{in} \cdot \kappa_m^{loss} \quad (41)$$

$$S_{m,t,y} = S_{m,t-1,y} + H_{m,t,y}^{in} + \sum_h Q_{h,m,t,y}^{dh} + \sum_b B_{b,m,t,y}^{dh} - Q_{m,t,y}^{gen} \quad (42)$$

$$Q_{m,t,y}^{gen} + N_{m,t,y}^{dh} = D_{m,t,y}^{heat} \quad (43)$$

$$\sum_h Q_{h,t,y}^{out} + \sum_b B_{b,t,y}^{ind} + N_{t,y}^{ind} = D_{t,y}^{heat,g} \quad (44)$$

$$\sum_h Q_{h,t,y}^{cind} = D_{t,y}^{cool,g} \quad (45)$$

Role: ensures thermally feasible operation and sector coupling consistency while preserving hourly heat/cooling service adequacy. The current v1.1 code additionally blocks cooling provision from electric-boiler variables and applies the CHP electric upper bound using surviving effective capacity rather than a purely static nameplate envelope.

7.11 11) Hydro Reservoir and Pumped Hydro Storage

Description: This block models reservoir and pumped-hydro storage over time. It enforces storage bounds, water-balance recursions with inflow and efficiency, feasible withdrawals (generation plus spill), and end-of-horizon storage targets. *Compact notation:*

$$\begin{aligned} \mathcal{U}^{dam} &\equiv \text{hdam_units}, \quad \mathcal{U}^{phs} \equiv \text{hphs_units}, \quad G_{u,t,y} \equiv \text{generation}_{u,t,y}, \\ S_{u,t,y} &\equiv \text{storage_level}_{u,t,y}, \quad P_u \equiv \text{max_load}_u, \quad O_{u,t,y} \equiv \text{sto_out_flow}_{u,t,y}, \\ F_{u,t,y}^{dam} &\equiv \text{hdam_inflow}_{u,t,y}, \quad \eta_u^{dam} \equiv \text{hdam_efficiency}_u, \quad \bar{S}_u^{dam} \equiv \text{hdam_max_storage}_u, \\ F_{u,t,y}^{phs} &\equiv \text{hphs_inflow}_{u,t,y}, \quad \eta_u^{phs} \equiv \text{hphs_efficiency}_u, \quad \bar{S}_u^{phs} \equiv \text{hphs_max_storage}_u. \end{aligned}$$

$$0.5 \bar{S}_u^{dam} \leq S_{u,t,y} \leq \bar{S}_u^{dam}, \quad u \in \mathcal{U}^{dam} \quad (46)$$

$$S_{u,1,y} = 0.5 \bar{S}_u^{dam} + F_{u,1,y}^{dam} \cdot P_u \cdot \eta_u^{dam} - \frac{G_{u,1,y}}{\eta_u^{dam}} - O_{u,1,y} \quad (47)$$

$$S_{u,t,y} = S_{u,t-1,y} + F_{u,t,y}^{dam} \cdot P_u \cdot \eta_u^{dam} - \frac{G_{u,t,y}}{\eta_u^{dam}} - O_{u,t,y} \quad (48)$$

$$\frac{G_{u,t,y}}{\eta_u^{dam}} + O_{u,t,y} \leq S_{u,t,y} + F_{u,t,y}^{dam} \cdot P_u \cdot \eta_u^{dam} \quad (49)$$

$$S_{u,T,y} \geq 0.5 \bar{S}_u^{dam} \quad (50)$$

$$0.5 \bar{S}_u^{phs} \leq S_{u,t,y} \leq \bar{S}_u^{phs}, \quad u \in \mathcal{U}^{phs} \quad (51)$$

$$S_{u,t,y} = S_{u,t-1,y} + F_{u,t,y}^{phs} \cdot P_u \cdot \eta_u^{phs} - \frac{G_{u,t,y}}{\eta_u^{phs}} - O_{u,t,y} \quad (52)$$

$$\frac{G_{u,t,y}}{\eta_u^{phs}} + O_{u,t,y} \leq S_{u,t,y} + F_{u,t,y}^{phs} \cdot P_u \cdot \eta_u^{phs} \quad (53)$$

$$S_{u,T,y} \geq 0.5 \bar{S}_u^{phs} \quad (54)$$

Role: preserves physically feasible hydro energy trajectories and prevents end-horizon reservoir depletion.

7.12 12) Ramping, Imports, RPS, and CO2 Policy

Description: This block combines short-term operational constraints (ramping and import smoothing) with strategic policy constraints (renewable portfolio share and annual CO2 cap). *Compact notation:*

$$\begin{aligned} \mathcal{U}^{disp} &\equiv \text{disp_units}, \quad \mathcal{U}^{rps} \equiv \text{nondisp_units} \cup \text{biomass_units} \cup \text{hydro_units}, \\ \mathcal{U}^{el} &\equiv \text{fossil_units} \cup \text{hphs_units} \cup \text{hdam_units} \cup \text{hrror_units} \cup \text{nondisp_units}, \\ G_{u,t,y} &\equiv \text{generation}_{u,t,y}, \quad I_{t,y} \equiv \text{imports}_{t,y}, \quad \bar{I}_t \equiv \text{import_ntc}_t, \\ r_u^{up} &\equiv \text{ramp_up_rate}_u, \quad r_u^{down} \equiv \text{ramp_down_rate}_u, \quad \rho_y \equiv \text{rps}_y, \\ E_y^{el} &\equiv \text{CO2_total}_y, \quad E_y^{ht} \equiv \text{CO2_total_heat}_y, \quad E_y^{ind} \equiv \text{CO2_total_industry}_y, \\ \bar{E}_y &\equiv \text{CO2_limit}_y, \quad \phi_u \equiv \text{CO2_factor}_u, \quad \phi_b^{th} \equiv \text{Thermal_CO2}_b, \\ \phi_f^f &\equiv \text{Fuel_CO2}_f, \quad B_{b,t,y}^{ind} \equiv \text{boiler_ind_generation}_{b,t,y}, \\ B_{b,m,t,y}^{dh} &\equiv \text{boiler_dh_generation}_{b,m,t,y}, \quad F_{f,t,y} \equiv \text{Fuel_ind}_{f,t,y}. \end{aligned}$$

$$G_{u,t,y} \leq G_{u,t-1,y} + r_u^{up} \cdot \text{CapEff}_{u,y}, \quad u \in \mathcal{U}^{disp} \quad (55)$$

$$G_{u,t,y} \geq G_{u,t-1,y} - r_u^{down} \cdot \text{CapEff}_{u,y}, \quad u \in \mathcal{U}^{disp} \quad (56)$$

$$I_{t,y} \leq \bar{I}_t \quad (57)$$

$$I_{t,y} \leq 1.2 I_{t-1,y} \quad (58)$$

$$I_{t,y} \geq 0.8 I_{t-1,y} \quad (59)$$

$$\sum_{u \in \mathcal{U}^{rps}} \sum_t G_{u,t,y} \geq \rho_y \left(\sum_{u \in \mathcal{U}^{el}} \sum_t G_{u,t,y} + \sum_t I_{t,y} \right) \quad (60)$$

In the implemented v1.2 code, the annual denominator is extended further by $\sum_t \text{no_load_elec}_{t,y}$, so unmet-electricity service also contributes to the policy base against which the renewable-share requirement is evaluated.

$$E_y^{el} = \sum_{u \in \text{fossil_units}, t} \phi_u \cdot G_{u,t,y} \quad (61)$$

$$E_y^{ht} = \sum_{b,t} \phi_b^{th} \cdot B_{b,t,y}^{ind} + \sum_{b,m,t} \phi_b^{th} \cdot B_{b,m,t,y}^{dh} \quad (62)$$

$$E_y^{ind} = \sum_{f,t} \phi_f^f \cdot F_{f,t,y} \quad (63)$$

$$E_y^{el} + E_y^{ht} + E_y^{ind} \leq \bar{E}_y \quad (64)$$

Role: preserves feasible system operation hour-by-hour while simultaneously enforcing portfolio and emissions policy trajectories.

7.13 13) Ramping Cost Linearization

Description: Auxiliary ramp-cost variables are lower-bounded by signed generation changes, yielding a linear representation of up/down ramping penalties in the objective.

$$\text{rampup_cost}_{u,t,y} \geq \text{ramp_cost}_u (G_{u,t,y} - G_{u,t-1,y}) \quad (65)$$

$$\text{rampdown_cost}_{u,t,y} \geq \text{ramp_cost}_u (-G_{u,t,y} + G_{u,t-1,y}) \quad (66)$$

Role: penalizes fast dispatch changes while preserving linearity.

8 Implementation Notes (Current Code Traceability)

- `Power_to_h2` is created with an extra loop over `flex_unit`; mathematically it is equivalent to one equation per (t, y) but instantiated repeatedly.
- `primary_hourly_reserve_down_units_minimum2` and `secondary_hourly_reserve_down_units_minimum2` use an up-reserve variable on the left-hand side.
- `CO2_cost_industry` is computed but not included in the final objective sum.
- With `Imp_price_inc=0.05`, the expression $(1 + \text{Imp_price_inc}/100)^n$ corresponds to 0.05% annual growth.
- The v1.1 code introduces explicit unmet-service variables `no_load_elec`, `no_load_heat_ind`, and `no_load_heat_dh`, together with the penalty terms `ceep_cost`, `no_load_cost`, and `no_load_cost_dh`.
- Input preprocessing in v1.1 migrates from SpreadsheetML/XML parsing to direct CSV/XLSX ingestion, while preserving the same identifier-matching logic across `year`, `period`, and unit or fuel labels.

- The current v1.1 script defines separate annual CAPEX dictionaries for electrolyzers, fuel cells, heat pumps, boilers, hydrogen storage, and stationary batteries instead of relying on a single generic flex-technology cost table.
- The present v1.1 code also sets `periods = list(range(1,10))`; treat this as an implementation-specific reduced horizon unless it is intentionally retained for the study design.
- The v1.2 workflow is launched from `main_rolling_horizon_v3.py` and dispatched through `scripts/modeling_approaches.py`. From there, the code calls either `Build_model.py` or `Build_model_RH.py`.
- The v1.3 workflow is launched from `main.py`. The wrapper `scripts/modeling_approaches.py` now manages the rolling-horizon loop directly and repeatedly executes the single multizone builder `Build_model.py`.
- Preprocessing in v1.3 reads `1_system_data.xlsx` to construct `systems`, `unit_system`, `N_From`, `N_To`, `transfer_limits`, and `link_check`, which then determine the multizone topology of the implemented constraints.
- The implemented builder adds explicit bilateral-flow and transmission-expansion objects, notably `imports_from` and `transfer_inv`, together with national-policy switches that aggregate regional emissions and renewable generation across systems.
- The bundled export stack for v1.3 is multizone and rolling-horizon specific: `export_plot_multizone_RH.py` recreates the scenario results folder and `save_all_results_to_csv.py` writes the internal `*_results` dictionaries to CSV for traceability.
- In the rolling-horizon branch, cumulative investment states are initialized, updated after each subproblem, and then reintroduced in the next subproblem as effective inherited capacity or investment carryover.
- Both v1.2 builders introduce the auxiliary variable `extra_cap_inv` and the penalty `extra_investment_cost`. This softens annual investment limits while keeping excessive expansion economically unattractive.
- The v1.2 implementation reactivates `spill_cost`, relaxes the hydrogen-storage minimum bound to direct non-negativity, and evaluates most continuous decision variables with explicit lower bounds equal to zero.

A Reserve Module Data Dictionary

This appendix consolidates the data dictionary entries used specifically in Sections 4.3 and 4.4.

A.1 Decision Variables for Reserve Modules

Variable (original model notation)	Type	Unit	Description
primary_hourly_reserve_up_units _{u,t,y}	Reserve	MW_e	Upward primary reserve from dispatchable units.
primary_hourly_reserve_down_units _{u,t,y}	Reserve	MW_e	Downward primary reserve from dispatchable units.
primary_hourly_stat_stoOUT_reserve_up_units _{b,t,y}	Reserve	MW_e	Upward primary reserve via stationary storage discharge margin.
primary_hourly_stat_stoOUT_reserve_down_units _{b,t,y}	Reserve	MW_e	Downward primary reserve via stationary storage discharge margin.
primary_hourly_stat_stoIN_reserve_up_units _{b,t,y}	Reserve	MW_e	Upward primary reserve via stationary storage charging margin.
primary_hourly_stat_stoIN_reserve_down_units _{b,t,y}	Reserve	MW_e	Downward primary reserve via stationary storage charging margin.
primary_hourly_FC_reserve_up _{c,t,y}	Reserve	MW_e	Upward primary reserve from fuel cells.
primary_hourly_FC_reserve_down _{c,t,y}	Reserve	MW_e	Downward primary reserve from fuel cells.
primary_hourly_ELY_reserve_down _{e,t,y}	Reserve	MW_e	Downward primary reserve from electrolyzers.
primary_hourly_ELY_reserve_up _{e,t,y}	Reserve	MW_e	Upward primary reserve from electrolyzers.
primary_hourly_HPind_reserve_down _{h,t,y}	Reserve	MW_e	Downward primary reserve from industrial heat pumps.
primary_hourly_HPind_reserve_up _{h,t,y}	Reserve	MW_e	Upward primary reserve from industrial heat pumps.
secondary_hourly_reserve_up_units _{u,t,y}	Reserve	MW_e	Upward secondary reserve from dispatchable units.
secondary_hourly_reserve_down_units _{u,t,y}	Reserve	MW_e	Downward secondary reserve from dispatchable units.
secondary_hourly_stat_stoOUT_reserve_up_units _{b,t,y}	Reserve	MW_e	Upward secondary reserve via stationary storage discharge margin.
secondary_hourly_stat_stoOUT_reserve_down_units _{b,t,y}	Reserve	MW_e	Downward secondary reserve via stationary storage discharge margin.
secondary_hourly_stat_stoIN_reserve_up_units _{b,t,y}	Reserve	MW_e	Upward secondary reserve via stationary storage charging margin.
secondary_hourly_stat_stoIN_reserve_down_units _{b,t,y}	Reserve	MW_e	Downward secondary reserve via stationary storage charging margin.
secondary_hourly_FC_reserve_up _{c,t,y}	Reserve	MW_e	Upward secondary reserve from fuel cells.
secondary_hourly_FC_reserve_down _{c,t,y}	Reserve	MW_e	Downward secondary reserve from fuel cells.
secondary_hourly_ELY_reserve_down _{e,t,y}	Reserve	MW_e	Downward secondary reserve from electrolyzers.
secondary_hourly_ELY_reserve_up _{e,t,y}	Reserve	MW_e	Upward secondary reserve from electrolyzers.
secondary_hourly_HPind_reserve_down _{h,t,y}	Reserve	MW_e	Downward secondary reserve from industrial heat pumps.
secondary_hourly_HPind_reserve_up _{h,t,y}	Reserve	MW_e	Upward secondary reserve from industrial heat pumps.

A.2 Parameters and Sets for Reserve Modules

Parameter / set (original model notation)	Category	Unit	Description
PrimaryReserve	Scenario flag	bool	Activates the primary reserve constraint block.
SecondaryReserve	Scenario flag	bool	Activates the secondary reserve constraint block.
res_inv	Scenario flag	bool	Activates reserve-investment related module switches.
reserve_factor_{<i>u,t,y</i>}	Technical	p.u.	Reserve-down scaling factor applied at the same hourly granularity as reserve variables (often time-invariant in data).
primary_reserve_units	Set	-	Index set for <i>u</i> in primary reserve variables primary_hourly_reserve_up_units_{<i>u,t,y</i>} and primary_hourly_reserve_down_units_{<i>u,t,y</i>} .
primary_reserve_StatStorage_units	Set	-	Index set for <i>b</i> in stationary-storage primary reserve variables.
primary_reserve_FC_units	Set	-	Index set for <i>c</i> in fuel-cell primary reserve variables.
primary_reserve_ELY_units	Set	-	Index set for <i>e</i> in electrolyzer primary reserve variables.
primary_reserve_HP_units	Set	-	Index set for <i>h</i> in industrial heat-pump primary reserve variables.
secondary_reserve_units	Set	-	Index set for <i>u</i> in secondary reserve variables secondary_hourly_reserve_up_units_{<i>u,t,y</i>} and secondary_hourly_reserve_down_units_{<i>u,t,y</i>} .
secondary_reserve_StatStorage_units	Set	-	Index set for <i>b</i> in stationary-storage secondary reserve variables.
secondary_reserve_FC_units	Set	-	Index set for <i>c</i> in fuel-cell secondary reserve variables.
secondary_reserve_ELY_units	Set	-	Index set for <i>e</i> in electrolyzer secondary reserve variables.
secondary_reserve_HP_units	Set	-	Index set for <i>h</i> in industrial heat-pump secondary reserve variables.